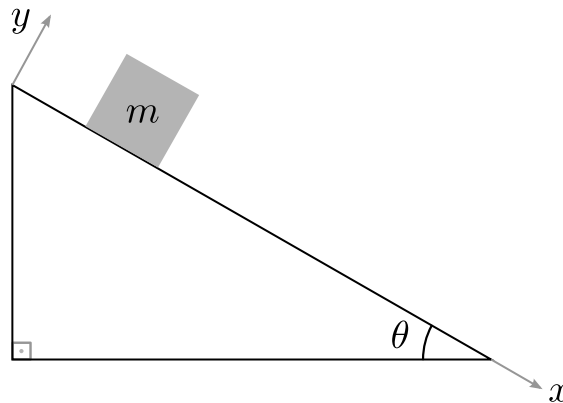


02MECHZ - Final
03.02.2023

Question 1

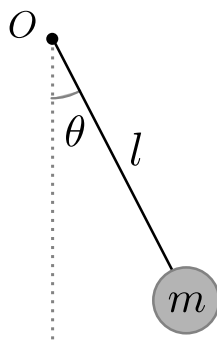


A block of mass m sliding down on an inclined plane with an inclination angle θ and kinetic friction coefficient μ_k

- (a) Draw the free-body diagram of the block.
- (b) Write down Newton's second law for y -direction in magnitude form in terms of N , m , g (gravitational constant), and θ .
- (c) Write down the magnitude of the frictional force acting in the block while it is sliding down in terms of μ_k , m , g , and θ .
- (d) Write down Newton's second law for x -direction in terms of $\ddot{x}$, g , θ , and μ_k .
- (e) Solve the differential equation (Newton's second law) you obtained in (d), and find $x(t)$ for the initial conditions $x(0) = 0$ and $\dot{x}(0) \equiv v(t) = 0$.
- (f) Calculate the position of the block at $t = 2$ s for $\theta = 30^\circ$ ($g \approx 10 \text{ m/s}^2$, $\sin 30^\circ = 1/2$, $\cos 30^\circ = \sqrt{3}/2$, $\mu_k = \frac{1}{2\sqrt{3}}$)

Options from (a) to (d) are each worth 0.2 points. Options (e) to (f) are each worth 0.1 points - Total: 1 point

Question 2



A simple pendulum of mass m and length l is swinging about its equilibrium point $\theta = 0$.

- (a) Draw the free body diagram of the mass.
- (b) What is the rotational inertia I with respect to the hanging point O in terms of m and l ?
- (c) What is the torque N acting on the mass with respect to point O in terms of l , m , g , and θ ?
- (d) First write down the Newton's second law for rotations in terms of N , I , and α (angular acceleration $\ddot{\theta} \equiv \dot{\omega} \equiv \alpha$). Then, plug the results you found in **b** and **c** into Newton's second law. (Note: Be careful about signs)
- (e) Solve the differential equation (Newton's second law) you found in **d** to obtain $\theta(t) = Ae^{i\omega t} + Be^{-i\omega t}$ for small angle approximation $\theta \ll 1$. (Givens: $\sin \theta \approx \theta$ when $\theta \ll 1$, solution assumption: $\theta(t) = e^{\lambda t}$)
- (f) Given that $\theta(t) = Ae^{i\omega t} + Be^{-i\omega t}$ can be written as $\theta(t) = C \cos(\omega_0 t)$, determine C for the initial condition $\theta(0) = \theta_0$.
- (g) By using $\theta(t)$ with the initial condition $\theta(0) = \theta_0$, calculate the angular velocity of the pendulum $\omega = \dot{\theta}(t)$. At what time t' , $\dot{\theta}(t)$ does take its largest value (the maximum speed of the pendulum) for the first time after being released from rest? By using $\dot{\theta}(t')$ calculate the maximum value of the kinetic energy of the pendulum.
- (h) Calculate the maximum kinetic energy of the pendulum from the point of view of conservation of energy: Assume that mass m has no potential energy at its equilibrium point ($\theta = 0$). First calculate the potential energy then the pendulum is at rest ($\theta = \theta_0$). We know that this potential energy will completely be converted to pendulum's kinetic energy when $\theta = 0$. (Given: $\cos \theta_0 \approx 1 - \theta_0^2/2$ for $\theta_0 \ll 1$)

Options from (a) to (f) are each worth 0.15 points. Options (g) to (h) are each worth 0.05 points -
Total: 1 point

Question 3

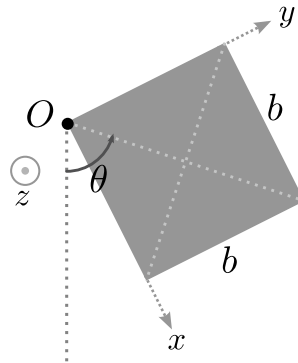
Find the force law $f(r)$ for a central-force field that allows a particle to move in a spiral orbit given by $r = k\theta^2$, where k is a constant. Given:

$$\frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) + \frac{1}{r} = -\frac{\mu}{l^2} r^2 f(r)$$

where l is the angular momentum and μ is the reduced mass.

Total: 1 point

Question 4



A homogenous square plate with side length b is hung from one of its corners O . The area density of the plate is $\rho_A = m/b^2$ where m is the total mass of the plate. We assume that the plate is effectively two dimensional, i.e., there is no mass distribution in z direction.

- Calculate the inertia tensor $[\mathbf{I}]$ by using $I_{ij} = \int \rho(r) (\delta_{ij} \sum_k x_k^2 - x_i x_j)$.
- Calculate the angular momentum vector $\mathbf{L}$.
- Calculate the magnitude of the torque N applied on the center of mass of the square plate.
- Write down Newton's second law for rotations.

Each option is worth 0.4 points - Total: 1.2 point